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Wearable Fall-Detection Systems: Sensors, Algorithms, and Energy trade-offs - A review

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ABSTRACT Falls are a leading cause of injury and death among older adults and workers. Smart wearable fall detection systems allows real-time monitoring and alerts, by combining sensing elements, embedded processing and wireless connectivity. This review analyses 12 recent prototypes (from 2021 to 2025), comparing them across sensors, type of device, algorithms, data transmission and power consumption. Most devices rely on IMU, and are placed in torso or waist; algorithms are simple thresholds, machine learning, or hybrid approach. This review documents the challenges of these studies, including the accuracy validity, power consumption trade-off and comfort and ergonomics. This review showed a power consumption relation between by the sensors and algorithms choice of the selected studies, where out-of-the-shelf boards with dedicated transmission modules consume less energy than the set with micocontrollers and separated boards. The trends on sensor location and algorithms were also approached, and some of the current limitations and some recommendations for future works were highlighted.

INDEX TERMS Fall detection, Wearable sensors, Human activity recognition, Smart monitoring system.

I. INTRODUCTION

Falls are a significant public health issue, ranking as the second main cause of accidental injury-related deaths worldwide, according to the World Health Organization (WHO). Each year, approximately 684,000 people die globally as a result of falls, with adults over the age of 60 accounting for the majority of these fatal incidents [1]. In Norway, the issue is particularly important. Nilsbakken et al. [2] found that elderly trauma patients are significantly more likely to receive under-triaged dispatches, with more than seven times the odds compared to other age groups. With projections indicating that by 2050, 22% of the global population will be aged 60 and above [3], Kinge et al. [4] estimate that Norway could face up to 39.45 billion NOK in expenditures related to unintentional injuries such as falls.

Beyond healthcare, falls are also a leading cause of occupational accidents. In the Norwegian construction industry, Winge et al. [5] showed that the most frequent accidents in the Norwegian construction industry are falls or contact with falling objects. Similarly, Holen et al. [6] demonstrated that falls are among the most common accident types on Norwegian fish vessels and fish farms. These findings emphasize the impact of falls across public and industrial sectors.

Both sectors can benefit from early fall detection systems based on remote monitoring. Such systems can help prevent complications associated with fall incidents and improve the efficiency of medical care. A fall detection system (FDS), in general terms, is an architecture designed to autonomously detect falls experienced by the user and notify caregivers as soon as these events occur.

FDSs can be mainly categorized into three types, based on the sensor architecture: vision-based systems, environmental systems, and wearable systems [7]. Vision-based systems rely on image processing techniques on the video frames or images captured by cameras around the region of interest. Artificial neural network (ANN) and tracking techniques can also be applied in order to increase the accuracy of the model [8]–[10]. Similarly, environmental systems rely on sensors placed in the environment, such as infrared [11], acoustic [12] or even a floor covered by piezoelectric sensors [13].

The main limitation of both type of systems is in the coverage area itself, as they demand sensor installation in constrained indoor spaces, typically within a room [14], where blind spots for detection may also occur.

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In contrast, in wearable-based systems the sensors used for fall detection are embedded within a wearable device that the user uses, such as a wrist band or a helmet. This dynamics makes these systems more attractive for end-users. Most of the wearable systems use inertial measurement unit (IMU) sensors, such as accelerometers, gyroscope, and magnetometer. They also usually contains a wireless communication module, facilitating the remote monitoring and integration of the tracking system into IoT (Internet of Things) networked platforms.

Some studies demonstrated that some of the main limitations of wearable systems are the rate of false positives [15], misleading sitting or quick bending as falls, comfort [16], as the user wears it all the time, and autonomy [17], since the continuous data acquisition, processing and transmission consumes a lot of energy.

To detect falls, various algorithms have been developed that analyze sensor data patterns, including acceleration, angular velocity, and orientation. These algorithms can be categorized in three types: threshold-based algorithms, machine learning-based algorithms and hybrid algorithms [18].

Threshold-based algorithms, are among the simplest and mostly used, by setting predefined limits on the inertial signals and triggering a fall detection when the signals pass a threshold. They can run on wearable device, however it can lead to a high number of false positives. Machine learning algorithms, on the other hand, analyze large volumes of historical data to identify complex patterns and features that indicate a fall, offering higher accuracy than threshold-based methods. However, due to the high computational load, run a machine learning algorithms on a wearable can be challenging, often running outside the device. Hybrid algorithms combine the strengths of both types, usually thresholding the signal that will be sent to the machine learning, improving its efficiency.

Previous studies reviewed different aspects of wearable FDSs, focusing on specific parts of the system. From the hardware side, Gutierrez et al. [19] studies vision-based fall detection systems while Subramaniam et al. focused on wearable systems, in different locations [20]. Regarding software and data analysis, Ramachandran et al. focused on FDSs that used machine learning algorithms [21] while Subramaniam et al. also focused on parameters and data analysis [20] and Casilari et al. focused analyzing the data quality of datasets for fall detection [22]. From the communication point of view, Villa et al. focused on FDS systems that used LPWAN technologies to transmit data with low consumption [23].

While the studies fulfill its primary goal, they mainly focus on isolated components, leaving the limitations out of the study goal and reaching similar conclusions.

This work aims to explore a wide variety of studies, demonstrate the current trends, regarding sensors, microcontrollers, position of the device, algorithms, alert systems, and autonomy. This will provide valuable insights for future works.

II. METHODOLOGY

The search for articles was conducted in well-known academic databases such as IEEE Xplore, MDPI, Scopus and Google Scholar. Since some information overlapped, this work aimed to identify studies that had not been considered in previous reviews.

The following aspects were selected as eligibility criteria for the studies:

- Studies published in peer-reviewed journals or presented at peer-reviewed conferences between 2021 and the first half of 2025.
- Studies that developed a working prototype.
- Studies that evaluates the accuracy and effectiveness of technologies in fall detection.
- Focus in at least one of the use-cases described in the previous subsection.

Additionally, some criteria were used to exclude studies from the selection, such as:

- Studies not written in English.
- Studies presenting devices other than wearables.
- Studies that do not send alarms when a fall is detected.
- Studies with poor accuracy.

The initial evaluation of the studies was performed by an examination of the titles and abstracts carried out by them. Studies that did not match with the research topics were discarded. All the researchers (us) performed independent reviews of the articles based on the criteria defined above.

At the end of of the filtering processes, 12 studies were selected for inclusion in this review.

III. OVERVIEW OF CONCEPTS

In the following sections, we describe some of the technologies applied in wearable FDSs.

A. SENSORS

Across wearable fall risk systems, the main sensors used are a 3-axis accelerometer and a 3-axis gyroscope. The accelerometer measures specific force, detecting the movement intensity. The gyroscope measures angular velocity, capturing the quick orientation changes that happen during a fall. Together describe most human motions during daily activities and falls, while being small, cheap and low power consumption. This combo is used in most research prototypes, being easy to integrate. Fusing the accelerometer and gyroscope allows for an higher reliability and performance compared to one sensor alone.

Other sensing options can be considered basing on the additional information needed in the context: barometers sense rapid pressure or altitude changes during a fall (e.g. going from standing height to floor level). Combining pressure and

accelerometer sensors allows the reduction of confusion with less false positive alarms [24]. Pressure/force sensors which converts mechanical strain to voltage in order to capture foot strikes and impact with the ground. Montanini et al [25] studied the application of this sensor underfoot or in-shoe, showing its effectiveness but placement-dependency. Finally, environmental and status sensors (e.g. gas, temperature, proximity, GPS) do not detect the fall but are useful for determining the context post-event conditions [26].

B. ALGORITHM

The three main types of algorithm for fall risk detection are: thresold based, machine learning and hybrid.

The thresold based use simple rules on IMU signals, typically involving a big spike in |a| followed with a rapid tilt change to declare a fall. It works by computing simple signals like the acceleration magnitude and orientation estimate, and then comparing the obtained value to fixed thresold. The most used pattern is impact and free fall: consists in checking the presence of a large |a| spike followed by a short window with very low |a|, plus a quick tilt change. Its possible to consider a short decision delay and posture check (e.g. if the person is lying after the impact) afterwards, in order to reduce false alarms. This method has the advantages of being low power consumption, easy to implement and run, while being not very suitable for more complex application due to its limited considerations of other factors. Its sensitive to where the device is worn, and for increasing its reliability adding safeguards (short decision delay, posture check, user cancel) is important.

The machine learning algorithm works by preprocessing the signals by applying noise cancellation filters such as low-pass filter in order to cancel noisy frequencies, then applying a window on the signal of a few seconds, extracting selected features, and running a classifier. Its advantages include high accuracy and precisely information about the type of fall. But labeled data and more compute and memory are needed, increasing the power consumption and lowering the autonomy.

The hybrid approach consists in a merge of the previous two. It works by detecting the fall using a simple thresold trigger, with a second confirmation step afterwards only if needed, consisting in Machine Learning check or other methods. Since the thresolds alone result fast but not highly reliable under complex conditions, and pure ML accurate but heavier, the hybrid approach keeps low power consumption and high accuracy by running the heavy approach only if needed, sending the alert after the confirm step.

C. DATA TRANSMISSION

Due to the advancements on the fall detection algorithms, the computational load for data processing exceed the wearable' capacity, specially for machine learning algorithms. This leads to solutions where the algorithm runs on another device, or on a server, requiring the addition of communication modules to the prototypes.

Today, a substantial amount of wearable FDSs proposed implements short-range transmission technologies, like bluetooth [23], Wi-Fi [27] or Zigbee [28], requiring the device receiving data to be in the vicinity. Smartphones are usually the target for bluetooth, and the user must carry at all times, a situation that is not always possible in all the application scenarios. with a Wi-fi module, on the other hand, the user can be further from the gateway, but it consumes more energy and it's unreliable in isolated areas.

Some works tried to explore middle and long-range transmission technologies applied in wearables FDS's, like LPWAN [23], 4G connection [29] and GSMs [30]. However those solutions tend to either consume more energy, like wi-fi, or makes the device heavier and more difficult to use.

The fall notification methods also vary a lot, from systems where a simple buzzer does the alarm, to SMS and cloud communications, ZigBee alert systems and other IoT implementations [31].

Table 1 provides a general comparison of these technologies, including range, frequency, band, and energy consumption.

Feature	Bluetooth	ZigBee	Wi-Fi	LoRa
Range	10 m	100 m	30 m	2-20 km
Frequency Band (Europe)	2.4 GHz	868 MHz	2.4 GHz	868 MHz
Data rate	1-3 Mbps	20-250 kbps	0.6-7 Gbps	0.3-50 kbps
Modulation	GFSK	BPSK, O-QPSK	OFDM, DSSS, FHSS	CSS
Transmission Power	Moderate	Low	Generally Higher	Low to Moderate
Power Draw	Low to Moderate	Very low	Relatively High	Very Low

TABLE 1. Comparison between Data Transmission Technologies [23], [32], [33]

In summary, LoRa and ZigBee can reach longer distances with low power. Bluetooth and Wi-Fi, however, have higher data rates, but also higher power consumption.

IV. ANALYSIS OF SELECTED STUDIES

This section presents a comprehensive analysis of the selected articles for this review. The key aspects of the studies, such as sensors used, fall detection algorithms implemented, data transmission architectures and autonomy are examined.

Table 2 summarizes the selected studies, highlighting the year, parameters investigated, performance and algorithm type, and autonomy.

Reference	Year	Sensors Used	Sensor Placement	Fall Detection Algorithm Type	Parameters Investigated	Transmission Technology	Autonomy (hours)	Observed Performance
[34]	2023	accelerometer	wrist	threshold-based statistical analysis	acceleration	-	2	90% accuracy
[35]	2024	accelerometer, gyroscope	waist	threshold-based, transformer neural network, self-attention	acceleration, angular velocity	Bluetooth	25	95.29% accuracy
[36]	2025	accelerometer	waist	CNN	acceleration	LoRa	178	96.67% detection rate
[37]	2021	accelerometer, gyroscope	waist (belt-buckle)	threshold-based	acceleration, angular velocity, tilt	Wi-Fi	-	90.83% accuracy
[38]	2025	accelerometer, gyroscope; tempera-ture/humidity; GPS	head (helmet)	threshold-based, free fall and impact	acceleration	GPS, Wi-Fi	2	-
[26]	2025	accelerometer, gyroscope; temperature; gas; proximity	head	threshold-based	acceleration, orientation	Wi-Fi	-	90% accuracy
[39]	2024	accelerometer, gyroscope; Pulse oximeter, temperature	torso (vest)	threshold based	accelerometer	Wi-Fi	-	-
[40]	2021	accelerometer, gyroscope	waist	deep-learning	acceleration, angular velocity	ZigBee	140	99.17% accuracy
[41]	2025	accelerometer, gyroscope; GPS	waist	threshold-based	accelerometer	Wi-Fi	-	97% accuracy
[42]	2023	accelerometer, gyroscope;	orthopedic walker	CNN	acceleration, angular velocity	Wi-Fi, Blue-tooth	-	97.4% accuracy
[43]	2022	triboelectric nanogenerator	shoe sole	CNN	pressure	Bluetooth	-	-
[44]	2022	accelerometer		CNN	acceleration	LoRa, Blue-tooth	53	95.55% accuracy

TABLE 2. Studies included in the systematic review

The selected studies in this review contained different focuses and novelties, as summarized on Table 3. This table highlights the main conclusions and limitations of each study

A. SENSOR SYSTEM

The sensor systems in the studied mainly utilize IMUs, with the triaxial accelerometer being the foundational component for detecting falls. Some systems also incorporate a gyroscope to capture rotational data, though many rely solely on acceleration to optimize for energy efficiency. The sampling rate for data acquisition varies, with common frequencies being 10 Hz, 20 Hz, and 50 Hz [35], chosen as a trade-off between capturing the dynamics of a fall and minimizing computational load.

Sensor placement significantly impacts performance. A general overview of the studied sensors can be found on Figure 1. The majority of the studies showed the waist or hip as the most effective location due to its proximity to the body's center of mass [36], getting better results than the wrist, which, despite its convenience, is more prone to generating false positives from daily activities. Salah et al. [44] focus on building a portable lightweight neural network for embeeded systems in areas with unstable internet, and besides describing the hardware system and sensors used,

Ref.	Focus	Contribution
[34]	Algorithm	Algorithm based on observations in the statistics of acceleration
[35]	Sensors and Algorithm	Embedded hybrid algorithm with neural network of lower parameters
[36]	Algorithm	Hybrid system that integrates a threshold-based model with two deep learning algorithms
[29]	Algorithm	A deep learning model trained by 1D CNN
[38]	Sensors and Transmission	FDS applied in construction sites in extreme heat conditions
[44]	Sensors and Algorithm	Proposes an architecture for wearable FDS reliable in areas with unstable internet
[26]	Sensors	Raspberry Pi based system with custom dashboard
[37]	Sensors and Algorithm	Design a threshold based classification method for a fall and a simple SMS notification system
[41]	Sensors and Algorithm	Lightweight wearable system for elderly
[40]	Algorithm	Combine deep neural network with CNN
[39]	Sensors	Low-cost ESP based wearable FDS

TABLE 3. Comparative insights and contributions of each study

the work does not explicitly informs the sensor location of the wearable system in the body.

Other studies innovated, on using state-of-the-art sensors for detecting variation of movement in shoe sole [45]. Despite the fact that different kinds of floors generate different signals, a machine learning algorithm can be used to correctly characterize each scenario.

Some proposed wearable systems could be used as part of the clothing, without direct touch to the body, like helmets [38], belts [37], vests [39] or even as an orthopedic walker [42].

B. FALL DETECTION ALGORITHM

The fall detection algorithms can be categorized into three main types: simple threshold-based methods [34], advanced deep learning models, and hybrid systems that integrate both [36]. Regardless of the type, the primary input is acceleration data, often processed into the Signal Vector Magnitude (SVM), with a common trigger being a peak acceleration

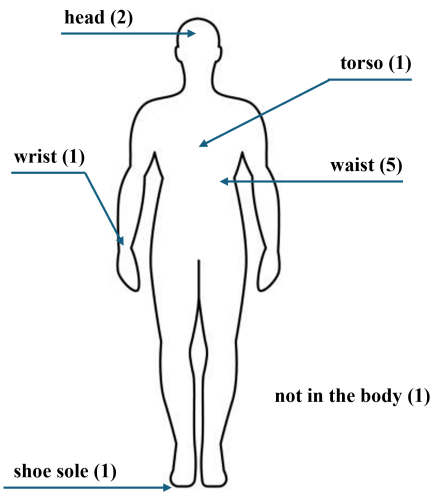


FIGURE 1. Distribution of wearable studies throughout the body with the count of devices in each area of the body

threshold between 2g and 3g [34]. Upon triggering, an algorithm analyzes a time window of data, typically lasting 1 to 4 seconds, to confirm the event. The performance varies with algorithmic complexity; threshold-based systems achieve accuracies around 85-90%, while hybrid and deep learning models report higher accuracies of 95-98% and sensitivities (recall) consistently above 96%. The execution location depends on computational demand. Lightweight algorithms, such as threshold-based triggers and optimized neural networks, can run entirely on the wearable device (edge computing) with inference times as low as 15 ms [35]. In contrast, more resource-intensive models like CNN-LSTMs are often executed on a remote cloud server, with the wearable device only transmitting data after a preliminary, on-device trigger, thereby conserving battery life.

C. DATA TRANSMISSION

Due to the wide variety of selected studies, some did not feature a well-developed data transmission unit. Li et al. [34], for example, presented a novel fall detection method but transmitted data to a computer via a USB-C cable. The majority of the studies, however, integrated short- and long-range transmission modules.

Among the technologies employed, Wi-Fi leads in preference, with six implementation, Bluetooth has 4 implementations, LoRa has two, Zigbee one, and one study has no data transmission implementation. Figure 2 summarizes these findings. In one of the studies, Salah et al. [44], directly compares their proposed system working with LoRa and Bluetooth, where LoRa presenting a higher battery life.

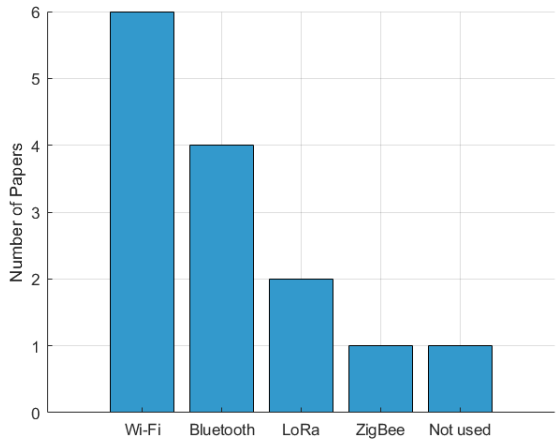


FIGURE 2. Distribution of data transmission technologies applied in the selected studies

The alarm systems varied significantly. Studies that implemented Bluetooth and Wi-Fi modules [35], [37], [38] designed the wearable to function primarily as a beacon, broadcasting the fall event to a smartphone. The smartphone application then forwards the alert to a designated emergency contact via SMS or e-mail.

Other applications adopted a three-node architecture: the wearable node, the gateway node, and the cloud node. In this setup, data sent from the wearable to the gateway is forwarded to the cloud, where computationally intensive machine learning algorithms can run efficiently and send real-time notifications to caregivers. This architecture is particularly effective for applications involving LoRa [36] and other technologies with limited data transfer rates.

This shows that the the selection of the transmission module influences not only in the range and latency of the system but also its scalability and suitability for continuous remote monitoring applications.

D. AUTONOMY

The majority of the studies in wearable for fall detection tend to focus on increasing the quality of data acquisition, with different sensors and parameters or in improving the accuracy of the fall detection algorithms, neglecting the autonomy of the system.

In a previous review focused on low power transmission devices, that took energy consumption as one of the analyzed metrics, only 4 from 20 studies analyzed explicitly showed a system autonomy analysis [23].

The different focus of the selected studies offers an overview of which factors influence more on the autonomy of the system. Half of the 12 selected studies presented energy consumption results, grouped on Table 4.

To estimate the average energy consumption in studies that lacked this information, the datasheets of the components used in each prototype were consulted. The batteries were estimated based on the calculated consumption, with the 1000 mAh being a simple 3.7V Li-Po battery and the 5000 mAh being a 5V power bank. Only one study could not be estimated, since the authors characterized the sensor [43].

Ref.	Algorithm Type	Components Consumption	Battery Capacity	Autonomy
[34]	Threshold	224 mA	120 mAh	2 hours
[35]	Threshold + TNN	16mA	400 mAh	25 hours (continuous)
[36]	CNN	24.3 mA	1000 mAh	178 hours (sleep mode)
[29]	deep learning	108.9 mA	1000 mAh	140 hours
[44]	CNN	37.72 mA		53 hours
[26]	Threshold	574.4 mA*	5000 mAh**	8.7 hours*
[37]	Threshold	126.64 mA*	1000 mAh**	8 hours*
[38]	Threshold	210.4 mA*	5000 mAh**	23 hours*
[41]	Threshold	529.4 mA*	5000 mAh**	9 hours*
[40]	deep learning	27 mA*	600 mAh**	140 hours*
[39]	threshold	77 mA*	1000 mAh**	13 hours *

TABLE 4. Energy consumption of selected studies. *Values estimated based on components datasheet. **The Battery capacity is estimated based on the available commercial batteries.

The choice of components has a crucial impact on both energy consumption and system autonomy. Among the microcontrollers, the studies with the highest documented or estimated current consumption used devices from the ESP family, specifically the ESP32 [34], [38] and ESP8266 [37]. This high consumption is primarily due to their embedded Wi-Fi connectivity. In contrast, using a microcontroller from the ARM family combined with a separate transmission module results in lower energy consumption than a single ESP module.

For example, using the Arduino Nano 33 BLE Sense microcontroller together with the LoRa RYLR988 communication module consumes less energy than the ESP32 alone [36], [38], [46]. The studies with the lowest energy consumption employed Arduino boards paired with dedicated transmission modules.

This shows that the use of out-of-the-shelf integrated microcontrollers might not be the optimal choice for an energy efficient wearable.

V. DISCUSSION

A review of the literature shows a clear correlation between wearable technology, edge computing and artificial intelli-

gence in the creation of fall detection systems. The dominant trend is the integration of sensors, mainly accelerometers, into discreet everyday objects such as wrist bands, watches [34], smart helmets [38], and even shoes [43]. This strategy of embedding technology into familiar objects, increases users willingness to apply them. At the same time, there is a significant shift from cloud computing to edge computing, enabling real-time analysis directly on the wearable device [35], [44]. This approach, supported by the development of energy-efficient algorithms, such as specialized deep neural networks (DNNs) and transformers [35], [40], is crucial for providing immediate alerts while conserving battery life [36]. The application of this technology is also expanding from its traditional focus on elderly care, to the safety of workers in hazardous environments [38], [39]. Taken together, these patterns indicate that the future of fall detection lies in specialized, energy-efficient, and intelligent edge devices that provide real-time, context-aware monitoring.

Most prototypes have been tested in controlled environments or with young volunteers, limiting their applicability to real-world conditions where factors such as body variability, movement patterns, and environmental conditions differ significantly. The comfort and usability of wearable devices play a decisive role in ensuring that users, especially older adults, wear them consistently. Integrating fall detection with broader health monitoring and emergency response systems can also improve reaction time and care coordination. Therefore, interdisciplinary collaboration among engineers, healthcare professionals, and ergonomics specialists will be essential to transform current prototypes into reliable, comfortable, and clinically validated solutions for real-world deployment.

VI. CHALLENGES

Despite promising results, FDSs still face several critical challenges that negatively impact their overall adoption and real-world effectiveness. One of the primary issues is in the limited generalizability of current validation studies. Many prototypes are tested on small, homogeneous groups and often young, healthy individuals, in simple daily applications, like sitting, which does not accurately reflect the characteristics of the elderly population or workers in different industries, the primary target users. Saod et al. [41] showed 97% accuracy using only 10 healthy adults, while Liu et al. [29] presented a significant drop in accuracy from 94.38% to 81.63% when the test group shifted from younger to older adults.

The sensor hardware limitations also constitute another challenge. While most systems uses IMUs, the placement location of these sensors affects detection accuracy. The studies whose devices were located on the wrist showed more likelihood to false positives due to regular hand movements, while waist or torso devices presented more stability, but could compromise user comfort. Some studies also looked to other sensors, like GPS, temperature, gas, pressure, in an attempt to increase accuracy and reliability, however this can lead to more complex algorithms and a higher computational capacity demand.

Another problem is the equilibrium between battery and computational power: keeping all the computational calculus on-device can drain the battery fast. On the other hand, streaming or sending data to the cloud can be better for the power, but this depends on the trasmission used (e.g. Wi-Fi/ZigBee). Li et al. with FD-DNN (Deep Neural Network), the vest sensor could last 140 h in low-duty mode, but only $>$30 h when streaming data more often (every 0.8 s), showing how communication cadence influence autonomy.

User comfort was an unexplored factor for most of the studies. Many prototypes do not account for the daily wearability needs of elderly users or workers in physically demanding environments. Devices such as helmets or vests may be too bulky or uncomfortable for long-term use. In Raghunath et al. [47], the smart-helmet prototype moved the battery into the user's pocket, in order to avoid exceeding weight, increasing comfort.

VII. FUTURE RESEARCH RECOMMENDATIONS

Energy efficiency and system autonomy is one of the biggest challenges and still is often neglected by current wearable FDSs. Many of the reviewed studies failed on providing system consumption and autonomy.

User comfort and device ergonomics also remain underexplored. Sensor placement has a huge impact on data quality and detection accuracy. Future studies should investigate alternative wearable, such as smart textiles, in-shoe sensors, or helmet-integrated systems, that balance accuracy with comfort.

Current systems often operate in isolation, limiting their impact on emergency response and long-term care. Future work should explore interoperability with electronic health records, cloud-based analytics platforms, and caregiver notification systems.

Exploring hybrid detection algorithms and integrating the transmission technologies with technologies like 5G and edge computing could further improve system performance.

Finally, future studies should aim on conduct tests in more users, giving a broader view on the reliability of the system, as well as scalability of the prototype.

VIII. CONCLUSION

This review provided a comprehensive analysis of recent developments in wearable FDSs, focusing on sensor technologies, algorithm types, communication architectures, and energy consumption. The review highlights that even though most of works utilizes IMUs, they tend to place the wearable device in the torso and waist area, closer to the center of mass.

It was also shown that to balance energy efficiency with data transmission, is crucial to use separate modules rather than out-of-shelf integrated boards. The choice of microcon-

troller and transmission module has a huge impact in the system's energy efficiency and autonomy.

The analysis revealed key limitations, particularly regarding energy efficiency, system autonomy, and user comfort, many prototypes still lack validation in real-world conditions or with older adult populations, the primary target users of such technologies. Additionally, aspects such as user comfort, ergonomics, and long-term usability were often not addressed, even though they are crucial for consistent use in daily life. Future research should focus on conducting large field tests, with different users and explore the scalability of the solutions.

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